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SOAL 1

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hal 1

1. SHA 256

NIM + Nama : 231112048Jordanniel (19 karakter = 152 bit)

a. proses menghasilkan w16 dan w18

Langkah 1 - Konversi pesan ke biner dan padding

panjang pesan L = 152 bit. Tambahkan bit 1, lalu k bit 0 (K = 295) lalu panjang pesan 64 bit (= 152), maka total 512 bit (1 blok)

Langkah 2 - Pecah jadi 16 word 32-bit (w0 - w15) :

w0 : 32333131	w4 : 65656c80	w8 : 00000000	w12 : 00000000
w1 : 31323034	w5 : 00000000	w9 : 00000000	w13 : 00000000
w2 : 384a6f72	w6 : 00000000	w10 : 00000000	w14 : 00000000
w3 : 64616e6e	w7 : 00000000	w11 : 00000000	w15 : 00000098

(w0 - w4 dari karakter 231112048J, dst ; w4 berisi padding bit 1 diakhir data ; w15 = panjang pesan 152 = 0x98)

Langkah 3 - Message Schedule

$$W_t : \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} \pmod{2^{32}}$$

dgn :

- $\sigma_0(x) = ROTR^7(x) \oplus ROTR^{18}(x) \oplus ROTR^3(x)$
- $\sigma_1(x) = ROTR^{17}(x) \oplus ROTR^{19}(x) \oplus SHR^{10}(x)$

* Hitung w16

$\sigma_0(w1) = \sigma_0(31323034) = e2492e2a$

$\sigma_1(w14) : \sigma_1(00000000) : 00000000$

$W16 = \sigma_1(w14) + w9 + \sigma_0(w_1) + w_0$ = 00000000 + 00000000 + e2492e2a + 32333131

w16 = 147c5f5b

* hitung w17

$\sigma_0(w2)$ $\sigma_0(384a6f72)$ = 78a55722

$\sigma_1(w15) = \sigma_1(00000098) = 005f0000$

$w17 = \sigma_1(w15) + w10 + \sigma_0(w2) + w1$ = 005f0000 + 0 + 78a55722 + 31323034

w17 = aa368756

* Hitung w18

$\sigma_0(w3) = \sigma_0(64616e6e)$: 8bdf7609

$\sigma_1(w16) = \sigma_1(147c5f5b)$: a943f7a6

$w18 = \sigma_1(w16) + w11 + \sigma_0(w3) + w2$: a943f7a6 + 0 + 8bdf7609 + 384a6f72

w18 : 686ddd21

Forte

5. Iterasi I = 0 sampai I = 2

Nilai Hash awal H0 - H7 :

H0 : 6a09e667	H3 : a54ff53a	H6 : 1f83d9ab
H1 : bb67ae85	H4 : 510e527f	H7 : 5be0cd19
H2 : 3c6ef372	H5 : 9b05688c	

Register kerja awal : a, b, c, d, e, f, g, h : H0 .. H7

Rumus tiap iterasi t :

- $\Sigma_1(e)$: ROTR^6(e) ⊕ ROTR^{11}(e) ⊕ ROTR^{25}(e)
- Ch (e,f,g) : (e AND f) ⊕ (NOT e AND g)
- Temp 1 : h + Σ_1(e) + Ch (e,f,g) + Kt + Wt
- Σ_0(a) : ROTR^2(a) ⊕ ROTR^{13}(a) ⊕ ROTR^{22}(a)
- Maj (a,b,c) : (a AND b) ⊕ (a AND c) ⊕ (b AND c)
- Temp 2 = Σ_0(a) + Maj (a,b,c)
- a' = Temp 1 + Temp 2, b' = a, c' = b, d' = c, e' = d + Temp 1,
  f' = e, g' = f, h' = g

Iterasi t = 0 (K0 = 428a2f98, w0 = 32333131) :

* Σ_1(e) : 3587272b , Ch(e,f,g) = 1f85c98c
* Temp 1 : 5be0cd19 + 3587272b + 1f85c98c + 428a2f98 + 32333131
  : 25ab1e99
* Σ_0(a) : ce20b97e, Maj (a,b,c) : 3a6fe667 → temp 2 : 08909ac5
* a : 2e3bb97e, b : 6a09e667, c : bb67ae85, d : 3c6ef372,
  e : cafb15d3, f : 510e527f, g : 9b05688c, h : 1f83d9ab

Iterasi t = 1 (K1 : 71374491, w1 = 31323034)

* Σ_1(e) : 98db5ac8, Ch (e,f,g) = 510e7a5f
* temp 1 : 1f83d9ab + 98db5ac8 + 510e7a5f + 71374491 +
  31323034 = 5bd72397
* Σ_0(a) : ae9a673a, Maj (a,b,c) : 2a2bae67 → temp 2 = d8c61sa1
* a = 399d3938 , b = 2e3bb97e, c : 6a09e667 , d = bb67ae85,
  e : 98461709, f = cafb15d3, g : 510e527f, h : 9b05688c

Iterasi t = 2 (K2 : ~~b5c0~~ b5c0fbcf, w2 : 38906f72)

* Σ_1(e) : e4599952, ch (e,f,g) : c99a5377
* Temp 1 : 9b05688c + e4599952 + c99a5377 + b5c0fbcf + 389a6f72
  = 36b9bb96
* Σ_0(a) : b0020a75, Maj (a,b,c) : 2919b97e → temp 2 : de1bc3f3
* a : 19d07f89, b : 399d3938, c : 2e3bb97e, d : 6a09e667,
  e : f21c6a1b, f : 98461709, g : cafb15d3, h : 510e527f.

Forte

Hasil akhir setelah iterasi $I = 2$

H_0 : 14d07f89	H_4 : f21c6a1b
H_1 = 349d3938	H_5 : 98961709
H_2 = 2e3bb97e	H_6 = cafb13d3
H_3 = 6a09e667	H_7 = 510e527f

$H_0 \ldots H_7$: 14d07f89349d39382e3bb97e6a09e667f21c6a1b98961709
cafb13d3510e527f